

Statistical Analysis Plan for COVID-19 Data on the ISARIC Data Platform

This document details the plan for an analysis of COVID-19 patient data submitted to the ISARIC database. Details of this cohort can be found in the MedRxiv reports available at <https://www.medrxiv.org/content/10.1101/2020.07.17.20155218v15> and the Data Descriptor paper available at <https://www.nature.com/articles/s41597-022-01534-9>. Copies of analysis plans from ISARIC Partners are available [here](#).

All contributors to the ISARIC COVID-19 database are invited to participate in this analysis through review and input on the statistical analysis plan and resulting publication. The outputs of this work will be disseminated as widely as possible to inform patient care and public health policy, this will include submission for publication in an international, peer-reviewed journal. ISARIC aims to include the names of all those who contribute data used in this analysis as cited collaborators on the resulting publication, subject to the [ISARIC publication policy](#).

Title of proposed research
External validation of a bleeding risk prediction model in COVID-19 patients receiving extracorporeal life support using the ISARIC dataset
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Research Plan

Background, Research Objectives, Scientific Value
Hemorrhagic complications are common and clinically consequential among critically ill patients receiving extracorporeal life support (ECLS), including extracorporeal membrane oxygenation (ECMO), contributing to morbidity, increased resource utilization, and mortality [1-3]. Although several bleeding prediction models have been proposed, their transportability across countries, healthcare systems, and practice patterns remains uncertain, limiting clinical adoption. We previously developed a machine-learning model to predict bleeding risk during ECLS in a multicentre Japanese cohort using routinely available laboratory and clinical variables (manuscript currently under review). Building on this work, the current project aims to externally validate a streamlined version of the model using laboratory measures that are broadly and consistently captured across international datasets, together with key clinical covariates available in the ISARIC case report forms. This study will provide evidence on model performance across diverse settings, assess the extent to which the model is generalizable outside the original development cohort, and inform future academic research on risk-stratified monitoring and management during ECLS.

¹ At least one lead must be based in an institution in an LMIC. If you would like to be connected with a potential partner researcher, please let us know at data@isaric.org.

Proposed Target Population
Adult (≥18 years) patients in the ISARIC COVID-19 dataset who received extracorporeal life support (ECLS/ECMO) during hospitalization. We will include confirmed or clinically suspected COVID-19 cases recorded in ISARIC and restrict analyses to patients with ECLS documented in the Treatments/Interventions domain.
Clinical Questions/Descriptive Analyses
1. <i>What are the demographic and clinical characteristics of adult COVID-19 patients receiving ECLS/ECMO in the ISARIC dataset, overall and by geographic region?</i> 2-1. <i>What is the frequency of GI bleeding among COVID-19 patients requiring ECLS/ECMO, and how does it vary by region and illness severity indicators (e.g., vasopressor use, renal replacement therapy (RRT), and lactate where available)?</i> 2-2. <i>What is the frequency of any recorded bleeding event during hospitalization, where compatible bleeding variables are available, and how does it vary by region and illness severity indicators?</i> 3. <i>How do key laboratory parameters (e.g., platelet count, CRP, PT/INR, APTT, D-dimer where available) and major supportive therapies differ between patients with vs without GI bleeding?</i> 4. <i>How does the bleeding prediction model perform in the ISARIC cohort with respect to discrimination and calibration?</i> 5. <i>How do the observed frequencies of selected in-hospital outcomes, such as in-hospital mortality and thrombotic complications, vary across model risk strata?</i>
Planned Statistical Analyses, Methodology and Representation
Data preparation and cohort definition The study population will include adult patients (aged 18 years or older) with COVID-19 registered in the ISARIC dataset who received extracorporeal life support (ECLS) during hospitalization. The index time point (Day0) will be defined as the date of ECLS initiation. In the ISARIC dataset, the ECLS initiation date will be operationally determined as the earliest calendar date on which ECLS is recorded as present in the Daily Form. All analyses will be anchored to Day0. Predictors at each prespecified time point (Day0 through Day5) will be defined using the nearest available measurement within a prespecified time window, preferentially using values recorded prior to any bleeding event occurring on the same day when applicable. Data preparation will include harmonization of laboratory units and clinically informed plausibility checks. Primary outcome definition The primary outcome will be the first occurrence of gastrointestinal bleeding during hospitalization after ECLS initiation. This endpoint was selected pragmatically because, in the ISARIC/WHO CORE case report form, gastrointestinal haemorrhage is recorded as a specific in-hospital complication, and the completion guide defines it as bleeding originating from any part of the gastrointestinal tract, from the oropharynx to the rectum. Because major bleeding and ECLS-specific bleeding sites are not systematically captured across all sites and case report form versions, gastrointestinal bleeding is used here as a pragmatic standardized endpoint and does not represent the full spectrum of hemorrhagic complications. For prediction model validation, events occurring prior to each prediction origin will be excluded to avoid outcome misclassification where temporal information is available. Secondary outcome Secondary bleeding outcome: Any recorded bleeding event during hospitalization after ECLS initiation. This outcome will be identified from harmonized bleeding-related complication variables and other bleeding-related records, irrespective of anatomical site, where available. Mortality: Mortality will be defined using ISARIC outcome status and outcome dates where available. The primary mortality outcome will be in-hospital death prior to discharge. Where date information is available, time-to-event analyses will be performed using time from Day0 to death or discharge. Thrombotic complications: Thrombotic complications will be defined as a composite endpoint including pulmonary embolism, stroke, and myocardial infarction as recorded in the ISARIC complications variables. Where available, the date of first occurrence will be used for time-to-event analyses; otherwise, outcomes will be analyzed as binary in-hospital event indicators. Respiratory support and healthcare resource utilization outcomes: Clinical course and resource utilization outcomes will include duration of invasive mechanical ventilation, ICU or HDU length of stay, and duration

of ECLS support where available. Use of renal replacement therapy and vasopressors or inotropes will be summarized and used as proxies for illness severity and clinical characterization.

Descriptive and epidemiological analyses

Demographic and clinical characteristics will be summarized overall and by geographic region. Continuous variables will be presented as median with interquartile range or mean with standard deviation, and categorical variables as counts and proportions. The frequency of gastrointestinal bleeding among ECLS patients will be estimated overall and stratified by region and severity proxies, including vasopressor use, renal replacement therapy use, and lactate levels where available. Comparisons between patients with versus without bleeding will include key laboratory parameters and major supportive therapies recorded in the ISARIC forms. Standardized mean differences will be reported to quantify between-group differences.

Prediction window

ECLS initiation will define Day0. Day0 through Day5 will be treated as prediction origins. At each prediction origin, available predictors will be used to predict clinically relevant outcomes, including gastrointestinal bleeding, occurring from the subsequent day onward. Thus, predictors at Day0 will be used to predict outcomes from Day1 onward, predictors at Day1 will be used to predict outcomes from Day2 onward, and similarly predictors at Day5 will be used to predict outcomes from Day6 onward. Model performance will be summarized primarily by prediction origin time point.

Primary analysis: External validation of the prediction model

External validation will be performed using a maximum model that includes all predictors from the original development study that are also available in the ISARIC forms. The maximum model will include demographic variables, ECLS modality where available, major supportive therapies, and laboratory measurements spanning hematology, biochemistry, coagulation, and inflammation. To enhance clinical implementability and robustness to missing data, two prespecified simplified models will also be evaluated. Simplified model 1 will include platelet count, C-reactive protein, neutrophil count, lymphocyte count, prothrombin time or international normalized ratio, and D-dimer. Simplified model 2 will include platelet count, C-reactive protein, neutrophil count, lymphocyte count, and prothrombin time or international normalized ratio.

When missingness is substantial, reduced model specifications excluding highly missing variables will be evaluated, and complete-case analyses will be conducted for each model specification. Model performance will be assessed using discrimination and calibration. Discrimination will be evaluated using the area under the receiver operating characteristic curve and the area under the precision-recall curve. Calibration will be assessed using calibration-in-the-large, calibration slope, and calibration plots. Ninety-five percent confidence intervals will be estimated using bootstrap resampling at the patient level to account for repeated predictions per patient. If miscalibration is observed, simple recalibration using intercept and slope adjustment will be reported while preserving the original model structure. Clinical performance at prespecified thresholds will be summarized using sensitivity, specificity, positive predictive value, and negative predictive value.

Subgroup analyses

Model performance will be examined across prespecified subgroups including geographic region, sex, body mass index categories, ECLS modality, renal replacement therapy use, and chronic kidney disease status. Within each subgroup, discrimination and calibration metrics will be estimated with confidence intervals. Performance will be summarized in tables and visualized using forest plots. Given potential variation in outcome prevalence across strata, precision-recall based metrics will be reported alongside receiver operating characteristic metrics.

Sensitivity analysis for reporting bias

To evaluate potential heterogeneity in outcome reporting across sites, a facility-level restricted analysis will be conducted. Restricted cohorts will be constructed by including only sites reporting at least a prespecified minimum number of outcome events. Model performance will be re-estimated within each restricted cohort and compared with estimates from the full cohort. Multiple thresholds will be explored to balance analytic stability and cohort size.

Phenotype assignment and characterization

In this study, bleeding risk phenotypes (high-risk vs low-risk) will be assigned using laboratory data only, and treatment interventions and supportive therapies will not be used as inputs for phenotype allocation. With ECLS initiation defined as Day0, predicted bleeding risk probabilities will be calculated at each time point from Day0 through Day5 using available laboratory predictors included in the prespecified simplified models. Patients will be classified as high-risk or low-risk at each time point based on the predicted probability. Thresholds will follow those prespecified in the development study whenever feasible; if direct application is not possible, sensitivity analyses will evaluate alternative thresholds based on the distribution of predicted risks in the external cohort.

Because phenotype assignment may change over time, transitions in phenotype status from Day0 through Day5 will be visualized using Sankey diagrams to illustrate movement between high-risk and low-risk categories across consecutive time points. The proportion of patients classified as high-risk at each time point will also be summarized to describe temporal shifts in cohort composition.

Phenotype will additionally be treated as a time-varying covariate to evaluate its association with clinical outcomes. As phenotype assignment is updated across Day0 through Day5, associations with mortality, thrombotic complications, and healthcare resource utilization outcomes will be estimated using appropriate regression frameworks, including time-to-event models with time-varying covariates or discrete-time hazard models where feasible. Adjusted effect estimates with 95% confidence intervals will be reported. As a sensitivity analysis, phenotype assignment will be fixed at Day0 and the same outcome association analyses will be repeated to assess consistency with the time-updated approach.

Exploratory interaction analyses will assess whether phenotype–outcome associations differ across prespecified clinical strata, and whether phenotype at Day0 modifies the association between selected interventions and outcomes. To examine treatment effect heterogeneity, we will perform exploratory causal effect estimation for interventions such as prone positioning and anticoagulation therapy, stratified by Day0 phenotype. Interventions will be defined within a prespecified early observation window relative to the prediction origin (e.g., during Day0 to Day1). Confounding will be addressed using appropriate adjustment approaches, including propensity score-based methods and/or doubly robust estimation, and phenotype-stratified effect estimates will be reported.

Handling of missing data

Missingness will be quantified for each variable and summarized by region and site. For longitudinal predictors, missingness across Day0 through Day5 will be described by variable and time point. The primary analysis will apply a prespecified imputation rule aligned with the development study to ensure reproducibility. Static categorical variables will be imputed using the most frequent category and static continuous variables using the mean. For longitudinal variables, last observation carried forward will be applied first, followed by time-point specific mean imputation for remaining missing values. Predictor values will be preferentially drawn from measurements recorded prior to outcome occurrence to minimize information leakage.

References

1. Dalton HJ, Garcia-Filion P, Holubkov R, et al. Association of bleeding and thrombosis with outcome in extracorporeal life support. *Pediatr Crit Care Med*. 2015;16(2):167-174.
2. Nunez JI, Gosling AF, O'Gara B, et al. Bleeding and thrombotic events in adults supported with venovenous extracorporeal membrane oxygenation: an ELSO registry analysis. *Intensive Care Med*. 2022;48(2):213-224.
3. Chung M, Cabezas FR, Nunez JI, et al. Hemocompatibility-Related Adverse Events and Survival on Venoarterial Extracorporeal Life Support: An ELSO Registry Analysis. *JACC Heart Fail*. 2020;8(11):892-902.

Data Required

[\(CQ\) COVID-19 Follow-up Questionnaire Domain](#)
[\(DM\) Demographics Domain](#)
[\(DS\) Disposition Domain](#)
[\(HQ\) Healthcare Encounters Domain](#)
[\(IN\) Treatments and Interventions Domain](#)
[\(LB\) Laboratory Test Results Domain](#)
[\(MB\) Microbiology Domain](#)
[\(RP\) Reproductive System Findings Domain](#)
[\(RS\) Disease Response and Clinical Classification Domain](#)
[\(SA\) Clinical and Adverse Events Domain](#)
[\(SC\) Subject Characteristics Domain](#)
[\(SV\) Subject Visits Domain](#)
[\(VS\) Vital Signs Domain](#)